SAINT MICHAEL COLLEGE OF CARAGA

EXPLORING THE POTENTIAL FOR AI INTEGRATION IN LIBRARIES: READINESS AND EXPECTATIONS OF LIBRARIANS AND USERS

A Thesis Presented To
The Faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF LIBRARY AND INFORMATION SCIENCES

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SAINT MICHAEL COLLEGE OF CARAGA
College of Computing and Information Sciences

CERTIFICATE OF RESEARCH APPROVAL

This thesis entitled **"**Exploring the Potential for AI Integration in Libraries: Readiness and Expectations of Librarians and Users" presented and submitted by Trisha L. Macarayan, Cheryl H. Atad, Janna Marie A. Montilla, Rezin A. Montilla, and Rosemarie T. Vitualla in partial fulfillment of the requirements for the degree of Bachelor of Library and Information Sciences is hereby accepted and recommended for Oral Examination.

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ABSTRACT

TITLE:	Exploring the Potential for AI Integration in Libraries: Readiness and Expectations of Librarians and Users
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EXPLANATORY SEQUENTIAL MIXED METHODS DESIGN

I. OBJECTIVES

With the rapid advancement of artificial intelligence, it is important to assess how prepared both librarians and users are for AI integration in libraries and to understand their expectations. Most existing studies focus on either librarians or users, and are conducted outside the Philippines, especially outside Agusan del Norte. This underscores the need for a localized study that considers both groups in this region. This research will evaluate their technological, individual, and organizational readiness, as well as their expectations for how AI may affect service quality, user experience, and library operations. The findings will help institutions plan and implement AI-driven innovations in library services more effectively.

II. METHODOLOGY

This study will make use of Explanatory Sequential Mixed-Method Design, starting with a survey to see how prepared librarians and users are for AI integration in libraries. After gathering survey results, interviews will be conducted to explore their thoughts and expectations in more depth. This approach allows the researchers to understand not just the data, but also the stories and experiences behind it, leading to a clearer and more meaningful understanding of the topic.

III. FINDINGS

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

IV. RECOMMENDATIONS

XXXXXXXXXXXXXXXXXXXXXXXXXXXXX

KEYWORDS: **XXXXXXXXXXXXXXXXXXXXXXXXXXXXX**

DEDICATION

This thesis is wholeheartedly dedicated to our beloved parents, families, and loved ones whose unwavering support and encouragement sustained us throughout this journey.

To our mentors — adviser, instructor, teachers, and panelists — we thank you for sharing your wisdom, expertise, and guidance that shaped the success of this study.

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Above all, we dedicate this work to Almighty God, for granting us wisdom, perseverance, and the opportunity to accomplish this milestone. Without His divine guidance, none of this would have been possible.

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Artificial Intelligence is becoming increasingly common today, and it has been for quite some time already. With its advancements, AI has greatly improved people's way of life. It has become the tool that makes tasks easier, faster, and more convenient.

Omame and Alex-Nmecha (2020) stated that Artificial Intelligence enables computers to mimic human thought and behavior. Some of these are language processing, speech recognition, pattern recognition, and learning. Morever, Ex Libris (2019) affirms that these AI systems help computers learn and enhance their performance. This, in turn, benefits users even more.

The continuous growth of Artificial Intelligence (AI) has brought significant changes to many areas, including libraries. For centuries, it has been altering the traditional ways of serving to library users by creating new chances for growth and ideas (Hasan et al., 2024). Adding AI to libraries makes them more innovative, data-driven, and interactive.

Encyclopedia Britannica (n.d.) indicates that a library is a place where people keep collections of books that they can read or study. Likewise, the American Library Association (n.d.) defines library as a collection of resources in various formats. These resources aim to educate, inform, or entertain a wide audience.

Libraries have been a key source of information for students, researchers, teachers, and the public for centuries. They guard knowledge and culture. However, as the world changes and technology evolves, libraries must adapt to better serve their patrons. They need to move from

being just a source of information to a tech-driven platform for the digital age (Isiaka et al., 2024; Sanjeeva et al., 2025). This study focuses on this important shift.

Yusuf et al. (2022) argued that the use of Artificial Intelligence in libraries enhances user engagement. It lets patrons use one platform to track their needs and get personalised services. This makes information easy to find, accurate, and promotes cost-effective, user-focused services maximizing its resources.

In addtition, Hodonu-Wusu (2024) noted that AI can transform how libraries serve their communities. By processing large amounts of data, AI can predict patrons' information needs. This greatly enhances the library experience ensuring all users have equal access to information.

As libraries increasingly adopt technology, Artificial Intelligence (AI) offers new ways to improve services and operations. AI tools, such as chatbots and virtual assistants, can assist users in real time, while recommendation systems suggest relevant books or resources. Moreover, AI can automate tasks like cataloguing, managing metadata, and analyzing usage patterns to help libraries operate more efficiently (Hasan, 2025). Furthermore, effective integration requires training librarians to use these tools confidently and ensure that AI systems align with the library's needs. Libraries must also follow ethical guidelines to protect privacy, ensure fairness, and promote responsible use of AI (Islam, 2025).

Building on these opportunities and challenges, this study conceptualizes AI integration as the perceived and anticipated application of AI technologies within library services, rather than their actual implementation, development, or evaluation. It examines the readiness and expectations of librarians and library users regarding potential AI-supported functions, including automated information retrieval, virtual reference services, recommendation systems, and operational support tools. No AI technologies will be deployed, accessed, or utilized during the study. Data collection is limited to survey questionnaires and interviews designed to capture

participants' perceptions, experiences, and expectations in library contexts. Accordingly, the study is descriptive and exploratory in nature and does not involve direct technological intervention or system-level experimentation.

Despite AI's potential, many libraries—particularly in developing countries—face significant barriers to adoption. Challenges such as inadequate infrastructure, unstable electricity, lack of skilled staff, resistance to change, privacy concerns, and high costs often limit the benefits of AI in library services (Tella, 2020; Echedom & Okuonghae, 2021; Isiaka et al., 2024; Barsha & Munshi, 2023). Nevertheless, AI continues to grow in relevance as it can transform information access, enhance learning and research, and support better decision-making for users (Okunlaya et al., 2022). By understanding these challenges alongside librarians' and users' readiness, this study aims to provide insights for effective AI adoption, ultimately enhancing library services and supporting education and community development.

A recent study in Eswatini found that librarians see AI as a way to boost productivity, help with information retrieval, and improve user services. However, they feel unprepared to use it. They lack technical knowledge, face weak ICT infrastructure, and struggle with unreliable internet. They also worry about privacy and data security (Hlatshwako & Tsabedze, 2024).

In the Philippines, AI is becoming more common as the country moves toward digital transformation. A study showed that librarians recognise the benefits of AI in improving library services and learning. However, many need more training and experience with AI tools (Paiste & Siago, 2024).

A study of academic librarians in the Bicol Region Librarians Council (BRLC) revealed that librarians generally have positive feelings about using AI. Many are eager to incorporate it into their work. They are open to using AI to enhance services and learning. However, challenges remain, like a lack of real-world experience and technical training (Juego & Masalinto, 2025).

Many people are interested in integrating Artificial Intelligence (AI) in libraries because of its potential to improve service delivery, enhance information access, and support efficient library operations. However, there is still limited understanding of how librarians and library users—particularly in provincial settings—perceive and prepare for such technological changes. While existing international and Philippine studies highlight generally positive attitudes toward AI, most focus on urban institutions, national samples, or professional groups alone, often excluding library users and overlooking organizational and technological readiness at the local level (Paiste & Siago, 2024; Juego & Masalinto, 2025).

Despite the growing relevance of AI in library services, there remains a notable lack of empirical research conducted at the provincial level, particularly in Agusan del Norte. Existing Philippine studies tend to concentrate on metropolitan areas or regional librarian councils and do not fully capture the realities of libraries operating in mixed environments such as school, academic, and public libraries within the province. Agusan del Norte presents a unique context characterized by uneven technological infrastructure, limited access to advanced digital tools, and varying levels of digital awareness among librarians and users. Consequently, findings from national or urban-centered studies may not accurately reflect the readiness and expectations of library stakeholders in this locality.

This absence of localized evidence creates a significant gap in understanding whether libraries in Agusan del Norte are technologically, individually, and organizationally prepared for potential AI integration. Without context-specific data, library administrators and policymakers lack a reliable basis for planning ethical, feasible, and sustainable AI adoption that aligns with local capacities and user needs.

Assessing readiness is therefore crucial. The adoption of new technologies extends beyond the availability of tools and infrastructure; it also depends on users' and librarians'

understanding of AI, institutional support mechanisms, and willingness to embrace change (Tella, 2020; Hasan et al., 2024). In provincial settings such as Agusan del Norte, where resources and digital exposure vary widely, readiness assessment becomes even more essential to ensure that AI initiatives are realistic and responsive to actual conditions.

In response to this gap, this study seeks to assess the readiness and expectations of both librarians and library users in selected libraries in Agusan del Norte regarding the potential integration of Artificial Intelligence. Specifically, it examines technological, individual, and organizational readiness, as well as service-related, user-experience, and outcome expectations. By providing localized and evidence-based insights, the study aims to support informed decision-making, guide future AI-related initiatives, and contribute to the limited body of research on AI readiness in Philippine libraries, particularly at the provincial level.

1.2 Review of Related Literature

The rapid advancement of Artificial Intelligence (AI) has significantly impacted the operations and service delivery of libraries. As institutions centered around information, libraries must continually evolve alongside emerging technologies to improve accessibility, effectiveness, and the overall user experience (Patris et al., 2025).

The ability of libraries to incorporate AI depends significantly on the preparedness and expectations of both librarians and users. Readiness influences their capacity to embrace and utilize new technologies, while expectations denote how they view the advantages and results of such integration (Machado et al., 2024; Srinivasan & Priyanka, 2025).

Many factors can affect this topic. These include age, education level, library type, and librarians' years of experience. Users' education, access to technology, and how often they use

libraries also matter. This section looks at studies and literature that explore the demographics, readiness, and expectations of librarians and users about adopting AI technologies in libraries.

Technological Readiness

Technological readiness measures how well individuals, organisations, and communities can adopt and use new technologies. It involves accessing digital tools and infrastructure, learning to use them, enhancing digital literacy, and becoming familiar with information and communication technologies. When appropriately developed, technological readiness helps individuals and organisations boost productivity, improve services, and encourage growth (Adil et al., 2025; Dowdy, 2020). Rahmat et al. (2020) also state that it reflects a person's mental and emotional ability to accept and adapt to AI technologies.

Libraries have undergone significant changes recently, making it essential for librarians to prepare for digital transformation. Adil et al. (2023) highlight that constant software and platform changes can overwhelm professionals, leading to increased stress. Research in Nigeria reveals that many librarians struggle to utilize technology effectively for delivering high-quality information services. This highlights the importance of structured support and skill development (Tunmibi & Okuonghae, 2023).

Insufficient training is a significant barrier. For instance, university libraries face challenges with automation because staff often lack knowledge of new systems or feel uncomfortable using them (Mancao & Dequeto, 2025). Some professionals worry that new tools might jeopardise their jobs. Institutions do not always support professional growth, so individuals often bear the costs themselves (Owolabi et al., 2022; Spurava et al., 2021). Failing to address these gaps can have a direct impact on service quality (Saibakumo, 2021).

Future awareness is another key factor. It is the ability to foresee how technology will evolve. Librarians who can predict the impact of AI, machine learning, and big data are better able to use these tools in research and user support (Siddiqui, 2025). Professionals who look ahead stay relevant and ensure their services evolve with technology (Cox & Mazumdar, 2024; Rajkumar et al., 2024). Without this vision, engagement may decrease, and efficiency may suffer.

A study in the Philippines found that academic librarians in Caraga and Northern Mindanao have a good understanding of artificial intelligence and a solid knowledge of other technologies, such as cloud computing and big data. Their positive attitudes towards these tools make them more likely to use them regularly. Continuous learning and training are essential for keeping skills up to date and responding to digital changes (Maraon & Agcito, 2024). Acceptance of technology directly affects its use; professionals who find tools practical are more likely to adopt them effectively.

Despite the availability of innovations, many librarians still have knowledge and skill gaps, which hinder their full utilization. Libraries often acquire new technologies without clear instructions, forcing staff to learn on the go (Adjei, 2023; Dowdy, 2020). Insufficient training can worsen service quality and reduce user satisfaction (Hamad et al., 2020; Ukamaka & Kakiri, 2021). Understanding and positive perceptions of technology are crucial for meaningful adoption (Otubelu & Anunobi, 2020).

Ultimately, a library's ability to benefit from modern tools relies on its staff's readiness. Knowledgeable, adaptable, and forward-thinking librarians can transform service delivery, streamline operations, and enhance user experiences (Gholami et al., 2018; Sun et al., 2018). Without this preparation, institutions risk falling behind, leaving users underserved and libraries unable to meet changing demands.

Individual Readiness

Librarians have long been adapting to technological changes, and these changes continue to impact the field of librarianship. Automation and digitization have improved the efficiency and capacity of library services and workflows over the years. This has opened the door to new ways of managing information (Hervieux & Wheatley, 2021). As technology keeps changing, the rise of artificial intelligence (AI) brings both chances and problems. Librarians need to learn new skills and adapt to stay useful in the digital age.

Arora et al. (2020) highlighted the increasing implementation of AI in academic libraries, notably the incorporation of virtual reference chatbots intended to aid patrons with common inquiries. These AI tools can help make information services more efficient, but in order for them to work well, librarians need to be ready to understand, use, and evaluate these technologies. By getting to know these technologies better, librarians can offer better, more user-centered services that meet the needs of today's library users.

Zimmett (2020) discussed the use of virtual assistants in libraries, emphasizing their potential to assist patrons with basic inquiries. However, Zimmett also highlighted the importance of librarians understanding the capabilities and limitations of these tools to ensure that information is delivered accurately and ethically. This reinforces the notion that readiness involves more than merely having access to technology—it also requires developing the digital skills, confidence, and adaptability necessary for the responsible use of AI in library services.

Building on this idea, Jan et al. (2024) said that being ready to use new technologies like AI depends on many things, such as how the organization works, how much money it has, how big it is, how it handles and protects data, and how willing each person is to change. In the context of libraries, this means that the effectiveness of AI integration relies heavily on the personal

preparedness and willingness of librarians to utilize AI tools for improved service delivery. When library staff have the proper training and motivation to use AI, they can make library services more responsive, efficient, and available to everyone. Hence, it is important to know how ready librarians are, because that will affect how well AI can be used to improve library operations and user experiences.

Organizational Readiness

Organizational readiness is a key factor in how well libraries can use new technologies like Artificial Intelligence (AI). Readiness includes not just the technological infrastructure, but also the leadership of the institution, the culture of the organization, and the group's willingness to try new things. Weiner (2009) characterizes organizational readiness for change as a collective psychological state wherein members are dedicated to enacting change and assured of their collective competence. This viewpoint emphasizes that readiness encompasses both motivational and capability dimensions; organizations necessitate adequate resources alongside intrinsic motivation and confidence to execute technological change effectively.

The study "Impact of Organizational Maturity on Acceptance and Use of Technology Among Library and Information Science Experts in the Public Libraries of Mashhad" shows that the level of maturity of an organization has a direct effect on how well its members use technology. Mature organizations have clear ways for people to talk to each other, good leaders, and a culture of always getting better. All of these things help staff accept and use new technologies. These results show that an organization is ready for AI only when its structure and culture are mature. Well-established policies, skilled employees, and support from management all create a suitable environment for integration.

Likewise, the research Evaluation of Organizational Readiness in Information Systems Adoption emphasizes that readiness is a multifaceted construct that includes technological, human, and managerial preparedness. The authors contend that technology adoption may falter despite sufficient infrastructure if organizational culture and leadership do not foster innovation. This evidence underscores the imperative for libraries to assess not only technical capacity but also strategic alignment, staff engagement, and management commitment prior to the implementation of AI-based systems.

In the Philippines, Abrigo and Torres (2022) looked at how ready libraries were to adapt to a changing service environment as they moved to digital operations. Their findings show that being prepared means having a plan, getting leaders involved, getting support from the workforce, and coming up with new ways to provide services. Libraries that included both management and staff in the planning and use of new technologies were more flexible and sure of themselves. The study also found that larger, better-funded libraries were more ready than smaller ones, which suggests that readiness depends on the size and support systems of the institution.

Hamid and Nayan's (2018) case study of the National Library of Malaysia on knowledge management adoption also found that readiness is based on both staff perception (willingness) and institutional capability (ability). The authors contend that a conducive working environment, defined by efficient policies, training, and infrastructure, fosters innovation and enables successful technology integration. This viewpoint corresponds with the tenets of organizational readiness for AI, highlighting the significance of librarians' attitudes, institutional backing, and favorable work environments in maintaining technological transformation.

These studies collectively illustrate that organizational readiness for AI integration in libraries encompasses more than mere technical preparedness. It requires leaders who are committed, organized management processes, the right amount of resources, empowered staff,

and a culture that encourages new ideas. Institutions that encourage teamwork, provide thorough training, and make sure that their tech goals are in line with their mission are more likely to use AI successfully. These findings emphasize the importance of assessing both organizational and human aspects of readiness to guarantee effective and sustainable AI integration in libraries.

Service - Related Expectations

Libraries have always been known to exist to provide access to knowledge, support learning, and meet the information needs of their users. In the modern era, librarians are increasingly looking toward AI technologies to improve the quality and efficiency of the services they offer. Service-related expectations in this context refer to what users anticipate from the library—prompt and accurate assistance, easy access to resources, personalized support, and overall satisfaction in interacting with library services. Librarians are able to meet these expectations by using AI tools that can make everyday tasks easier, organize resources, and answer user questions more quickly (Kalbande et al., 2024).

However, the adoption of AI in libraries is not without challenges. Librarians frequently encounter challenges, including inadequate institutional support, insufficient funding, deficient infrastructure, and deficiencies in professional training, which can impede their capacity to fully execute AI-driven services (Ali et al., 2025). Even with these problems, librarians are still dedicated to making services better because they know that AI can make information easier to find and library interactions more efficient and enjoyable (Orubeb, 2025).

Research shows that AI can automate specific tasks, such as cataloging, making recommendations, or answering common questions; however, the human touch remains very important. People still expect librarians to help them find their way, explain things, and help them with complex research needs. AI is viewed as an enhancement to conventional services, aiding

librarians in better fulfilling user expectations rather than supplanting them (D'Souza, 2024; Das & Islam, 2021).

Additionally, reliability and responsiveness are part of service-related expectations. Students and other library users are using AI tools more and more to help them with their schoolwork, but they still value the accuracy, trustworthiness, and personal touch that librarians offer. Research indicates that users value AI's ability to enhance task efficiency, such as swiftly locating pertinent resources or obtaining prompt assistance. At the same time, librarians continue to be accessible for inquiries and to offer more profound insights (Sambrano et al., 2025).

To sum up, service-related expectations for AI-integrated libraries include the quality, speed, ease of access, and responsiveness of library services. Users want faster, more accurate, and more personalized services that improve their library experience, and librarians are expected to use and adapt AI tools to meet these needs. To meet these expectations, libraries must strike a balance between new technology and the expertise of librarians. AI should be used to improve library services, not replace them.

User-Experience Expectations

User-experience expectations refer to how librarians and library users expect AI to enhance their interaction with library systems in terms of ease of access, efficiency, and satisfaction. Scholars have emphasized that AI technologies in libraries can significantly enhance the personalization of services, thereby improving the overall user experience. For instance, Kumar (2024) highlighted that AI-enabled recommendation systems provide personalized suggestions, enabling users to discover relevant materials easily. Additionally, Yusuf et al. (2022) noted that AI integration can streamline search and retrieval processes, enabling faster and more accurate access to information, which aligns with users' expectations for improved service. Similarly, studies

by Korada et al. (2025) indicate that AI can increase accessibility for users, including those with disabilities, thereby making library resources more inclusive and user-friendly.

Furthermore, Prasanna (2025) discussed how AI-powered virtual assistants in libraries enhance navigation and inquiry responses, making them more interactive and engaging, thereby meeting user demand for a seamless digital experience. Mandal (2024) argued that users now expect AI to work in tandem with librarians to provide comprehensive and meaningful support—not to replace human interaction but to augment it. In a similar vein, Abudulsalami et al. (2025) found that user satisfaction increases when AI tools reduce wait times and complexity in accessing library services. Lastly, Hamam (2025) posited that AI's role in creating more intuitive library interfaces directly impacts user engagement and loyalty, reinforcing the critical role of user-experience expectations in technology adoption in libraries.

Outcome Expectations

Outcome expectations refer to the perceived benefits that librarians and users anticipate from AI integration, such as improved efficiency, better access to resources, and enhanced service delivery. Several recent studies have documented that AI adoption in libraries positively affects operational efficiency and user satisfaction. For example, Mahmud et al. (2023) identified that AI-powered automation in cataloguing and classification significantly reduces manual workload, allowing librarians to focus on higher-value tasks. Similarly, Haman (2024) reported that AI facilitates more efficient resource management and accelerates digital transformation initiatives within libraries.

Moreover, Baig and Yadegaridehkordi (2025) highlighted that outcome expectations among library users include increased productivity in research and academic tasks due to faster and more accurate information services enabled by AI. Hodonu-Wusu (2024) underlined the role

of AI in expanding equitable access to library resources by simplifying interfaces and overcoming language or accessibility barriers. Furthermore, Echedom and Okuonghae (2021) discussed that AI can improve decision-making quality by providing librarians with data-driven insights, aligning with institutional goals of enhancing service quality.

Additional studies, such as that by Okunlaya et al. (2022), emphasize the transformative potential of AI in reshaping library workflows to be more user-centric and innovative. Juego and Masalinto (2025) found that both librarians and users expect AI integration to foster greater engagement and satisfaction, facilitating seamless digital interactions. Lastly, Patris (2020) cautioned that while outcome expectations are high, realizing them depends on sufficient training, infrastructure, and organizational support, which remain areas needing improvement in many libraries.

1.3 Theoretical Framework

This study is grounded in four frameworks: the Technology–Organization–Environment (TOE) Framework by Tornatzky and Fleischer (1990), the Technology Readiness Index (TRI) by Parasuraman (2000), the Technology Acceptance Model (TAM) by Davis (1989), and Braun and Clarke's Thematic Analysis (TA) Framework (2012). The TOE, TRI, and TAM frameworks explain the variations in readiness and expectations of librarians and library users toward the adoption of Artificial Intelligence (AI) by emphasizing technological, organizational, and human factors. Braun and Clarke's TA framework supports the qualitative component of the study by providing a systematic approach to analyzing participants' perceptions and experiences related to AI integration in libraries.

The Technology–Organization–Environment (TOE) Framework developed by Tornatzky and Fleischer (1990) explains how organizations decide to adopt new technologies based on three

contextual factors: technological, organizational, and environmental. In this study, the TOE framework is used to examine the readiness of libraries for Artificial Intelligence (AI) integration by considering the availability of technological infrastructure, institutional support, leadership, policies, and external influences. This framework helps explain how internal and external organizational conditions affect libraries' capacity to adopt AI systems and supports the assessment of organizational and technological readiness among librarians and institutions.

Technology Readiness Index (TRI) proposed by Parasuraman (2000) focuses on the individual's preparedness to embrace and use new technologies. It measures readiness through four dimensions: optimism, innovativeness, discomfort, and insecurity. In the context of this study, the TRI explains the individual readiness of librarians and library users to engage with AI technologies. It highlights how personal attitudes, confidence, and perceived risks influence their willingness to use AI in library services, emphasizing that successful AI integration depends not only on available technology but also on users' and librarians' psychological readiness.

The Technology Acceptance Model (TAM) developed by Davis (1989) explains how individuals decide to accept or reject new technology based on perceived usefulness and perceived ease of use. In this study, TAM is used to understand librarians' and users' expectations and acceptance of AI in libraries. It explains how positive perceptions of AI's benefits and ease of operation can influence favorable attitudes and intentions to use AI-based library services. This model supports the analysis of service-related, user-experience, and outcome expectations associated with AI integration.

Braun and Clarke's Thematic Analysis Framework (2012) is used in this study to guide the qualitative phase of data analysis, particularly in interpreting the interview responses of librarians and library users. This framework provides a systematic approach to identifying, analyzing, and reporting patterns or themes within qualitative data through six phases: familiarization with the

data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report. In the context of this research, the TA framework supports a deeper understanding of participants' perceptions, readiness, and expectations regarding AI integration in libraries. It complements the quantitative findings by allowing the researchers to explore underlying meanings, experiences, and viewpoints, thereby strengthening the explanatory sequential mixed-method design of the study.

Together, the TOE, TRI, TAM, and Braun and Clarke's Thematic Analysis (TA) frameworks provide a comprehensive view of AI integration in libraries. The TOE framework focuses on institutional and technological readiness, the TRI emphasizes individual preparedness, and the TAM explains user acceptance and expectations of AI. Braun and Clarke's TA framework complements these models by guiding the systematic analysis of qualitative data to capture librarians' and users' experiences and perceptions. The integration of these frameworks enables the study to examine the technological, organizational, human, and experiential factors that shape readiness and expectations for AI integration in libraries.

1.4 Schematic Diagram

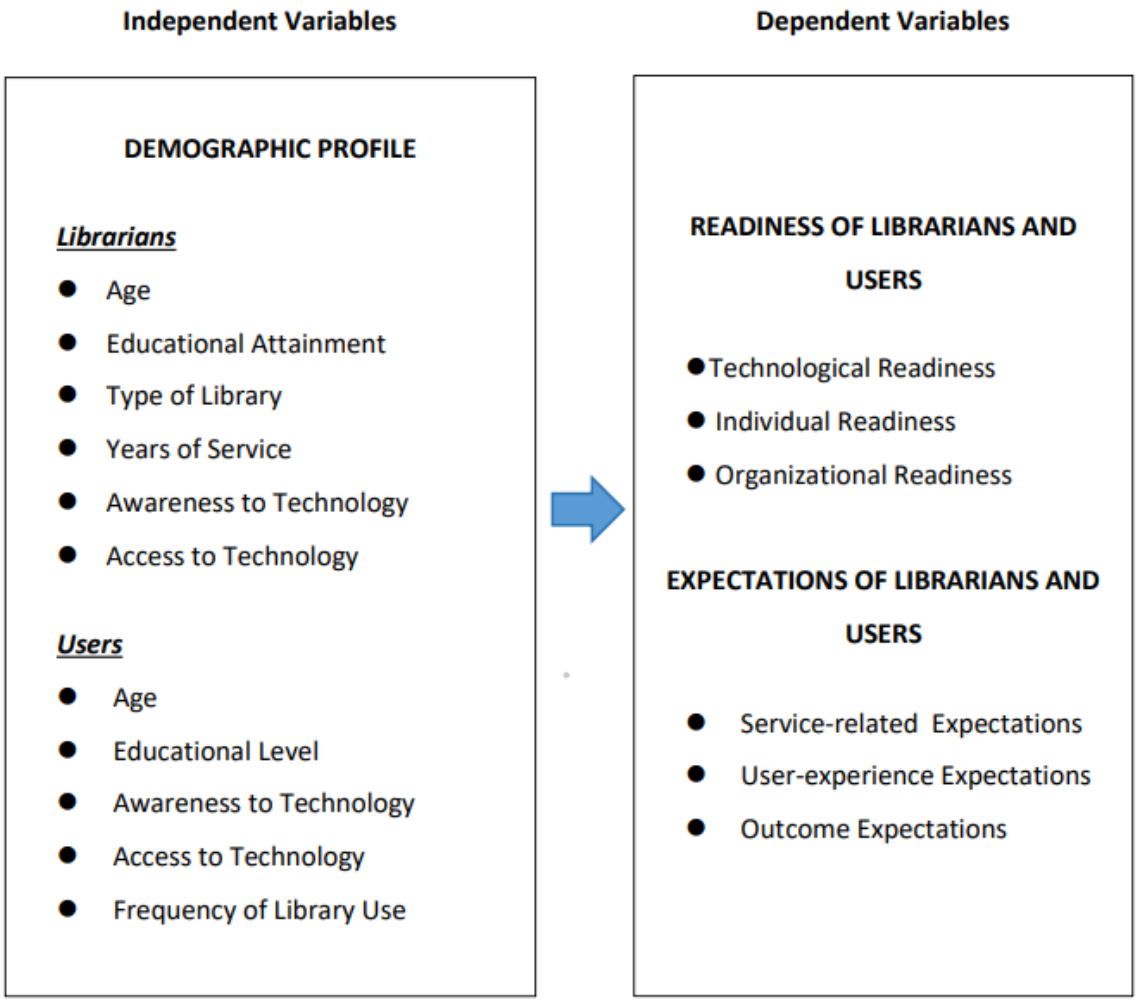


Figure 1. This schematic diagram interplays between the independent and dependent variables.

1.5 Statement of the Problem

This study aims to determine the readiness and expectations of librarians and users for the potential integration of Artificial Intelligence (AI) in libraries.

Specifically, it seeks to answer the following questions:

1. What is the demographic profile of the respondents in terms of :

1.2 Librarians

1.1.1 age;

1.1.2 educational attainment;

1.1.3 type of library;

1.1.4 years of service;

1.1.5 awareness to technology; and

1.1.6 access to technology?

2.2 Library Users

1.2.1 age;

1.2.2 educational level;

1.2.3 awareness to technology;

1.2.4 access to technology; and

1.2.5 frequency of library use?

2. What is the level of readiness of both librarians and library users to adopt Artificial Intelligence in terms of:

2.1 technological readiness;

2.2 individual readiness; and

2.3 organizational readiness?

3. What is the level expectations of both librarians and library users in terms of :
    - 3.1 service-related expectations;
    - 3.2 user-experience expectations; and
    - 3.3 outcome expectations?
4. Is there a significant relationship between the demographic profile of the respondents and their level of readiness to adopt AI in libraries?
5. What are the perceived benefits and risks do librarians and users identify in relation to AI adoption, particularly the ethical, technical, and professional issues?
6. What strategies or support mechanisms do librarians and users suggest to facilitate the adoption of AI in library services?

1.6 Hypothesis

This study aims to assess both the preparedness and expectations of librarians and library users for the potential integration of Artificial Intelligence (AI) in libraries. The research will take place during the academic year 2025–2026 in selected schools, academic, and public libraries in Agusan del Norte. Its purpose is to offer insights into how these institutions and their stakeholders might adapt to AI technologies, and to identify anticipated opportunities and challenges.

However, this study is limited only to selected libraries in Agusan del Norte. Thus, the findings may not be generalized to other locations. Data will be based on participants' perceptions and responses, which may be shaped by personal experiences or limited familiarity with AI. In addition, the research will not address the development or evaluation of AI systems, but will center exclusively on preparedness and expectations among librarians and users.

1.7 Definition of Terms

For this study, the following terms were operationally defined as follows:

Academic Library. Refers to the libraries in Agusan del Norte that are situated within a college or university, which support an institution's educational and research objectives by providing access to learning materials and digital resources.

Artificial Intelligence (AI). Pertains to a computer-based systems or tools that has the ability to perform tasks that usually require human intelligence, such as information retrieval, data analysis, and automated library services, as considered in this study.

AI Integration. The anticipated application of artificial intelligence in library services, focusing on librarians' and users' perceptions and expectations of potential AI functions without actual technological implementation.

Demographic Profile. Refers to the characteristics of the librarians and library users of the libraries in Agusan del Norte. This includes educational attainment, type of library, and years of service for librarians, and educational level, access to technology, and frequency of library use for library users.

Expectations. Refers to the expected changes or improvements that the librarians and library users of Agusan del Norte libraries have that are associated with the integration of AI in libraries.

Individual Readiness. Pertains to the personal willingness, confidence, and adaptability of librarians and library users of Agusan del Norte to engage with AI technologies in library operations.

Librarian. Denotes a professional librarian or a library-in-charge of the libraries in Agusan del Norte who is responsible for managing library resources and services, participating as a respondent in assessing AI readiness and expectations.

Library. Refers to an institution in Agusan del Norte that provides access to information resources, both physical and digital, serving as the setting for assessing AI readiness and expectations.

Library User. Refers to any student, faculty member, or individual who regularly accesses library services and resources of the libraries in Agusan del Norte, participating as a respondent in the study.

Organizational Readiness. Pertains to the extent to which the library in Agusan del Norte, as an institution that possesses the structure, support, and leadership necessary for adopting AI technologies.

Outcome Expectation. Refers to the perceived results or benefits that the librarians and library users of Agusan del Norte expect AI to bring, such as improved efficiency, better access to resources, or enhanced service delivery.

Public Library. Refers to a library funded by the government or local community in Agusan del Norte, providing information and educational services to the general public, and is included as part of the study's library types.

Readiness. Encompasses the overall preparedness of the librarians and library users of Agusan del Norte to adopt AI in terms of technological, individual, and organizational aspects.

School Library. Denotes a library in Agusan del Norte that is situated within a secondary educational institution that supports teaching and learning activities, which also serves as one of the study sites.

Service-Related Expectation. Refers to the anticipated improvements in library operations and services that the librarians and library users in Agusan del Norte have, that is associated with the adoption of AI tools.

Technological Readiness. Described as the availability of technological infrastructure, tools, and skills necessary for implementing AI in libraries of Agusan del Norte.

User-Experience Expectation. Pertains to how librarians and library users of Agusan del Norte expect AI to enhance their interaction with library systems, including ease of access, efficiency, and satisfaction.

CHAPTER 2

RESEARCH METHODOLOGY

2.1 Research Design

This study will use the Explanatory Sequential Mixed-Method Design. First, researchers will collect data using a modified questionnaire. They will measure how ready librarians and library users from selected libraries in Agusan del Norte are for AI integration. Then, the researchers will conduct interviews to understand their expectations and perceptions better. The qualitative data from the interviews will be analyzed using Braun and Clarke's Thematic Analysis, which involves identifying, analyzing, and reporting patterns or themes within the data. This approach is helpful because it interprets statistical results in light of participants' experiences. This leads to a deeper understanding of the research problem.

2.2 Research Materials and Methods

The questionnaire and interview guide that the researchers will use to collect the data to examine the readiness and expectations of librarians and library users in the potential integration of AI in libraries was modified from a previous study, "Perceptions of Artificial Intelligence: A Survey of Academic Librarians in Canada and the United States" by Wheatley & Hervieux (2021). With the modified research instruments, the researchers will then apply the Explanatory Sequential Mixed-Method Design in collecting the data.

To secure the accuracy and reliability of the research instruments, they will undergo content validation and a reliability test. Pilot testing will also be conducted to establish the reliability of the instruments.

2.3 Research Environment, Population and Sample

This study will be conducted in Agusan del Norte, one of the five provinces of the Caraga Region in Mindanao, Philippines. It is situated in the northwestern part of the Agusan Valley and has a total land area of approximately 2,730.24 square kilometers. The province is composed of ten municipalities and one component city, Cabadbaran City, which also serves as its capital. Agusan del Norte is known for its fertile plains, vast forestlands, and abundant river systems, including the Agusan River—one of the country's longest and most significant waterways. Its economy is mainly supported by agriculture, forestry, and fishing, with growing developments in commerce, technology, and education. The province is also a home to several educational institutions that contribute to its social and economic development.

The respondents of the study will be the librarians and library users from selected libraries in Agusan del Norte that meet the criteria established by the researchers. There were five criteria that the libraries must meet to be included in the study. The requirements include (1) Location, which indicates that the libraries must be located within Agusan del Norte, (2) Library Type, in which the library must be under the school, academic, or public library (3) Staffing / Librarian Qualification, this states that the librarian or library-in-charge must have atleast one year of experience in working in a library. (4) Technology Infrastructure, which states that libraries must have facilities and technological resources like computers, reliable internet connectivity, and access to digital tools. Moreover, lastly (5) Consent to Participate, which indicates that the library must be willing to participate in the study.

Table 1. Respondents of the Study as to Library Users

Respondents	Population	Sample	
	Freguency (*f*)	Freguency (*f*)	Percentage (%)
School Library			
Philippine Science High School - Caraga Region	400	62	19.08
Saint Michael College of Caraga (Basic Ed.)	468	72	22.15
Agusan National High School	200	31	9.54
Mindanao Institute - UCCP	20	3	0.92
Academic Library			
Asian College Foundation	10	2	0.62
Northern Mindanao School of Fisheries	25	4	1.23
Saint Michael College of Caraga (College)	66	10	3.08
Saint Joseph Institute of Technology	55	9	2.77
Father Saturnino Urios University (College)	700	107	32.92
Candelaria Institute of Technology Cabadbaran, Inc.	50	8	2.46

Public Library			
Cabadbaran City Library and Information Center	56	9	2.77
Butuan City Library	50	8	2.46
Total	2100	325	100.00

Table 1 displays the distribution of the respondents among the library users who will be included in the study.

2.4 Research Data and Analysis Plan

To collect data for this study, the researchers first wrote to the PLAI-Caraga Region Librarians Council. They contacted the secretary to request a list of accredited libraries in Agusan del Norte.

Next, the researchers applied selection criteria to identify which libraries qualified for the study. They then sent approval letters to the presidents and vice presidents of academic affairs at the chosen libraries, asking for permission to conduct research in their institutions.

The researchers will use the Explanatory Sequential Mixed Method Design. This approach combines quantitative and qualitative methods in two phases. In the first phase, they will gather quantitative data through survey questionnaires for librarians and library users in the selected libraries. This will assess their readiness and expectations for integrating Artificial Intelligence (AI) into libraries. The second phase will include qualitative interviews with selected respondents to explain and explore the quantitative results in more depth.

2.5 Research Ethical Standards

The researchers in this study understand that credible scholarly work requires ethical integrity. Fairness, transparency, and respect for human dignity will guide every phase, from data collection to analysis and distribution. They will follow accepted ethical standards to ensure their actions reflect integrity, responsibility, and social accountability. Protecting participant rights, intellectual property, and confidentiality, while ensuring voluntary participation, are key to the research process. These ethical guidelines help advance knowledge and safeguard the welfare and trust of library patrons and librarians.

A. Protection of Intellectual Property Rights (IPR)

Adhering to Philippine copyright law for all materials produced, including research instruments and analytical frameworks, would be a top priority for these researchers to protect their intellectual property rights. There will be proper citations and acknowledgments for all secondary data and other sources used. Since developing new logos, emblems, and/or brand identities is not part of the study, trademark registration is not required.

B. Informed Consent

Librarians and users in the study will receive a clear explanation of its goals, methods, possible risks, and expected benefits. Before joining, they will get a consent form in simple language. This form will outline their rights, including the right to leave the study at any time without any negative effects. Since the study does not involve animals or organisms, there are no additional ethical concerns about animal welfare.

C. Data Privacy and Confidentiality

The researchers will follow the rules of the Data Privacy Act of 2012 (RA 10173) in the Philippines very closely to protect the private information of the people who will take part in the study. We will only collect basic demographic information, such as age and gender, and not names or other identifying information. All information gathered will be anonymized to protect participant privacy. Physical records will be stored in a secure location that is only accessible by the research team, and digital information will be stored in password-protected folders. This will ensure that all participant information is safe and protected. Any data shared for research or academic purposes will stay completely anonymous, and it will only be shared with participants' explicit permission and in strict accordance with ethical and legal standards.

D. Voluntary Participation and Freedom to Withdraw

The researchers will make sure that everyone who takes part in this study does so willingly. The freedom to withdraw from the study at any moment without consequence will be made abundantly evident to participants. Participants will not be pressured in their decisions. The hiring process will be transparent and honest. This study does not involve animals or non-living creatures. So, no extra ethical considerations are needed.

E. Minimization of Harm and Risk Management

While the study poses minimal risk to participants, potential psychological, reputational, and academic risks are recognized. Psychological risks may include mild discomfort or anxiety when participants reflect on their knowledge, readiness, or concerns regarding artificial intelligence in library services. Reputational risks may arise if participants fear that their responses

could be associated with their professional role or institution, while academic risks may include concerns that participation or responses could affect academic standing or employment.

To minimize these risks, participation will be strictly voluntary, and participants will be informed of their right to withdraw from the study at any time without penalty. All survey and interview questions will be presented in neutral and non-judgmental language, and no personally identifiable information will be collected. Data will be anonymized and reported only in aggregated form, and participants will be assured that the study is conducted solely for academic purposes and does not evaluate individual or institutional performance. As the study does not involve physical intervention, animals, or biological materials, the 3Rs principle (Replacement, Reduction, and Refinement) is not applicable.

F. Beneficence and Contribution to Knowledge

This study aims to contribute to the improvement of library services and academic research by identifying the readiness, expectations, and concerns of librarians and library users regarding AI integration. While participants may not receive direct or immediate personal benefits, their insights will help raise awareness of technological needs, training gaps, and ethical considerations in library environments.

For participants, involvement in the study provides an opportunity to reflect on their experiences and perspectives related to emerging technologies in libraries. Participants may also request access to the study's findings, allowing them to understand how their responses contributed to the overall results.

For institutions, the findings will offer evidence-based information that may support decision-making, policy development, and strategic planning related to responsible AI adoption in libraries. The study also contributes to the limited local literature on AI readiness in libraries in

Agusan del Norte, serving as a useful reference for future researchers, library administrators, and educators.

G. Justice and Fair Participant Selection

The study will use clear, fair rules on who can and cannot participate to ensure everyone is treated fairly and honestly. We will ensure that people from diverse backgrounds are included in the study and will be treated fairly. This study will not involve animal trials, so the choice of species will not be relevant.

H. Data Integrity and Accuracy

The researchers will choose participants carefully. They will use clear criteria for who can join and who cannot. They will put fairness and objectivity first at all times. We will ensure everyone is involved, so no one is left out or treated differently due to age, gender, ethnicity, or any other personal trait. Since the study does not involve animal trials, species selection is not applicable.

I. Transparency and Honesty in Reporting

The researchers are committed to disseminating the study's findings transparently and candidly. There will be no changes or selective reporting of any kind; all results will be shown exactly as they are. The researchers will ensure that all copyrighted or patented materials in the study are credited. This protects intellectual property rights and maintains academic integrity.

J. Use of Patented or Copyrighted Materials

The researchers in this study will not use any patented technologies, so they do not need permission from patent holders. We will properly cite any copyrighted materials, such as software, books, or other resources. If required, the proper permissions will be obtained to make sure that all copyright laws are followed and that research is done in an ethical way.

K. Considerations for Human Subjects

The researchers make sure that each study participant knows the goals, methods, risks, and benefits before they agree to take part. Participation is entirely voluntary, and anyone can leave at any time without facing any repercussions. Since no animals are used in this study, ethical approval for animal welfare is not required.

L. Ethical Clearance and Institutional Approval

The researchers are fully committed on carrying out the study in accordance with the highest level of ethical conduct. Before collecting any data, we will ask the ethics review board for approval. Recording the approval date will help ensure that all actions follow ethical guidelines. This way, we prioritise the rights and well-being of everyone in the study.

M. Responsible Use of AI and Other Technologies

To make the study more effective and organised, the researchers will use AI tools and software. These will help with data management, information analysis, and literature review. Every output from the AI will be monitored and evaluated to keep the study's integrity. Researchers will ensure accuracy and guard against any misrepresentation of results.

CHAPTER 3

RESULTS AND DISCUSSION

DELETE BOX AFTER READING

Check the guidelines in research writing.

Make sure to integrate Mendeley in the citation.

Use Calibri 11 and double space

3.1 Presentation and Analysis of Data

3.2 Proposed Intervention/ Output

CHAPTER 4

SUMMARY, CONCLUSION, RECOMMENDATION,

DELETE BOX AFTER READING

Check the guidelines in research writing.

Make sure to integrate Mendeley in the citation.

Use Calibri 11 and double space

4.1. Summary

4.2. Conclusion

4.3.Recommendation

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: 2025 - present

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: 2025 - present

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
none	**none**

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: 2020-2021
: **Barobo National High School (Senior High)**
: P-1B Townsite, Barobo, Surigao del Sur
: 2022-2023
Tertiary : **Saint Michael College of Caraga**
: Atupan St., Brgy. 4, Nasipit, Agusan del Norte
: 2025 - present

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
none	**none**

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Personal Information

Sex : Male
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Height : 178 cm
Weight : 58 kg
Civil Status : Single

Educational Attainment

Elementary : **Kinayan Elementary School**
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: 2016-2017

Secondary : **Barobo National High School (Junior High)**
: P-1B Townsite, Barobo, Surigao del Sur
: 2020-2021
: **Barobo National High School (Senior High)**
: P-1B Townsite, Barobo, Surigao del Sur
: 2022-2023

Tertiary : **Saint Michael College of Caraga**
: Atupan St., Brgy. 4, Nasipit, Agusan del Norte
: 2025 - present

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
none	**none**

APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo
B - Instrument with Informed Consent
C - Map of the Research Locale *(Include Coordinates)*
D- Narrative & Photo Documentation
E- Ethics Committee Approval

CERTIFICATE OF TECHNOLOGY ASSESSMENT

CERTIFICATE OF AUTHENTIC AUTHORSHIP

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